

Lecture 6:

AI504 - Knowledge Representation

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Canonical models

For each set of sentences Γ we have an associated cononical model $\mathcal{M}_\Gamma$

Take $\Gamma = \{(a, c), (b, d), (b, c), (c, b), (d, c), (d, e)\}$

Where $P = \{a, b, c, d, e\}$

Lets construct a new model where the domain is the signature P . (the signature is the entire universe)

$$\mathcal{M}_\Gamma = (P, \llbracket \cdot \rrbracket, P \rightarrow \mathcal{P}(P))$$

In this model $\llbracket p \rrbracket = \{x : \Gamma \vdash (x, p)\}$

For this example then.

$$\begin{aligned}\llbracket a \rrbracket &= \{x : \Gamma \vdash \text{All } x \text{ are } a\} = \{a\} \\ \llbracket b \rrbracket &= \{x : \Gamma \vdash \text{All } x \text{ are } b\} = \{a, b, c, d\} \\ \llbracket c \rrbracket &= \{x : \Gamma \vdash \text{All } x \text{ are } c\} = \{a, b, c, d\} \\ \llbracket d \rrbracket &= \{x : \Gamma \vdash \text{All } x \text{ are } d\} = \{a, b, c, d\} \\ \llbracket e \rrbracket &= \{x : \Gamma \vdash \text{All } x \text{ are } e\} = \{a, b, c, d, e\}\end{aligned}$$

Properties of canonical models

The canonical model only satisfy senteces, which Γ proves. I.e., for any sentence ϕ ,

$$\Gamma \vdash \phi \iff \mathcal{M}_\Gamma \models \phi$$

e.g. In other model $\mathcal{N}$, if $\mathcal{N} \models \Gamma$ then i must satisfy any sentece that Γ proves (by soundness).

But $\mathcal{N}$ could also satisfy other sentences too, not $\mathcal{M}_\Gamma$. This only satisfies the senteces whcih Γ proves.

Proof of completeness

Assume $\Gamma \models \phi$.

This means that every model satisfying Γ must satisfy ϕ . Since $\mathcal{M}_\Gamma$ satisfies Γ by definition. Then

$$\mathcal{M}_\Gamma \models \phi$$

We know that $\mathcal{M}_\Gamma$ only satisfies sentences which Γ proves. Since $\mathcal{M}_\Gamma \models \Gamma$, then

$$\Gamma \vdash \phi$$

□

Soundness & Completeness

Says $\vdash$ and $\models$ agrees. I.e., for any Γ, ϕ then

$$\Gamma \vdash \phi \iff \Gamma \models \phi$$

Explanation

For any Γ and ϕ , either $\Gamma \vdash \phi$ (there exists a proof), or $\Gamma \not\vdash \phi$ (there exists a counter-model. i.e a model which satisfies Γ but falsifies ϕ)

Algorithmic question

Given Γ and ϕ , either

- find a proof of ϕ from Γ

or

- find a model satisfying Γ but not ϕ